Journal of Clinical Anesthesia 111 (2026) 112161

Contents lists available at ScienceDirect

Journal of Clinical Anesthesia

journal homepage: www.elsevier.com/locate/jclinane

Original Contribution

Association of target-controlled versus manually controlled infusion with postoperative recovery outcomes in elderly cardiac surgery patients: an international multi-cohort study

Yun-Xiao Bai, M.D [a,1], Xia Li, Ph.D., M.D [b,1], Shi-Qiang Chen, M.D [a], Zhen-Zhen Xu, Ph.D., M.D [a], Yan-Ting Wang, Ph.D., M.D [a], Fang-Ling Zhou, M.D [a], Qing-Ping Wu, Ph.D., M.D [a,*]

[a] *Department of Anesthesiology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430022, China*
[b] *Department of Anesthesiology, Renmin Hospital of Wuhan University, Wuhan 430060, China*

HIGHLIGHTS

- TCI optimizes drug delivery, but its impact on postoperative recovery in elderly cardiac surgery patients is unclear.
- In this multicohort study, TCI improved hemodynamic stability, reduced reintubation rates, and shortened ICU stays.
- TCI may enhance postoperative recovery and clinical outcomes in elderly cardiac surgery patients.

ARTICLE INFO

Keywords:
Postoperative recovery
Target-controlled infusion
Cardiac surgery
Hemodynamic stability

ABSTRACT

Background: Target-controlled infusion (TCI) optimizes drug delivery and hemodynamic stability using individualized pharmacokinetic models. We compared the effects of TCI versus manually controlled infusion (MCI) of propofol and remifentanil on postoperative recovery in elderly cardiac surgery patients.
Methods: An international retrospective cohort study was conducted using data from Union Hospital and the INSPIRE dataset. Elderly patients (≥65 years) receiving TCI were matched 1:1 with those receiving MCI using propensity scores to adjust for confounding variables. The primary outcome was the length of stay in the ICU. Secondary outcomes included postoperative complications and intraoperative hemodynamic stability.
Results: Of 1398 enrolled patients, propensity score matching yielded 468 in the Union cohort and 270 in the INSPIRE cohort. In the Union cohort, TCI was associated with a significantly shorter ICU stay (Mean Difference [MD] -0.92 days; 95% CI: −1.68 to −0.17; $P = 0.017$), lower odds of reintubation (OR 0.94; 95% CI: 0.90–0.99), reduced AKI incidence (OR 0.84; 95% CI: 0.78–0.91), and lower in-hospital mortality (OR 0.95; 95% CI: 0.92–0.99) compared with MCI. Validation in the INSPIRE cohort confirmed shorter ICU stays (MD -1.20 days; 95% CI: −2.32 to −0.80; $P = 0.036$) and lower reintubation rates (OR 0.96; 95% CI: 0.92–0.99). In both cohorts, TCI was associated with superior intraoperative hemodynamic stability, characterized by lower blood pressure variability and a reduced burden of hypotension.
Conclusion: TCI is associated with superior intraoperative hemodynamic stability, significantly shorter ICU length of stay, and improved postoperative recovery profiles compared with MCI in elderly cardiac surgery patients.

1. Introduction

Cardiovascular diseases are the leading cause of death globally, with approximately 2 million patients undergoing cardiac surgery worldwide annually [1]. As life expectancy increases, the patient demographic is shifting toward older individuals with multiple comorbidities, increasing surgical complexity and the risk of postoperative complications [2,3]. This highlights the need for advanced perioperative

* Corresponding author at: Department of Anesthesiology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430022, China.
E-mail address: wqp1968@163.com (Q.-P. Wu).
[1] These two authors contributed equally to this work and share first authorship.

https://doi.org/10.1016/j.jclinane.2026.112161
Received 9 December 2025; Received in revised form 19 January 2026; Accepted 17 February 2026
Available online 4 March 2026

strategies to optimize recovery and improve clinical outcomes in this vulnerable population [4,5].

Target-Controlled Infusion (TCI) is an anesthetic delivery technique designed to maintain stable, pre-set plasma concentrations of anesthetic agents based on pharmacokinetic models [6,7]. Unlike manual infusion (MCI), TCI allows anesthesiologists to titrate drug delivery dynamically, potentially improving intraoperative hemodynamic stability by mitigating blood pressure fluctuations and variability in anesthetic depth. While the pharmacokinetic advantages of TCI are well-documented, its impact on postoperative recovery profiles in elderly cardiac surgery patients remains less explored [8–10]. This is clinically significant because intraoperative hemodynamic instability, such as prolonged hypotension, is strongly linked to postoperative complications including acute kidney injury, prolonged mechanical ventilation, and reintubation, all of which extend the length of stay in the intensive care unit (ICU) [11,12].

The specific aims of this study were two-fold. First, we aimed to compare the effects of TCI versus MCI on postoperative recovery profiles, with a primary focus on the length of ICU stay. Second, we assessed the impact of TCI on intraoperative hemodynamic stability. We hypothesized that TCI is associated with superior intraoperative hemodynamic stability, and consequently, improved postoperative recovery outcomes in elderly patients undergoing cardiac surgery.

2. Methods

2.1. Study design and participants

This multi-cohort retrospective observational study included patients aged 65 and older scheduled for elective cardiac surgery (CABG, valve, aortic surgery, or combined procedures) requiring cardiopulmonary bypass (CPB) in China and Korea (Fig. 1). Patients meeting any of the following criteria in either cohort were excluded: (1) use of any inhalational anesthetic agents used during surgery; (2) administration of TCI or MCI for only one anesthetic agent (either propofol or remifentanil) rather than both; (3) requirement for mechanical ventilation support before surgery; (4) concomitant non-cardiac surgeries; (5) died or were discharged within 48 h postoperatively; (6) missing relevant clinical data.

The primary cohort was collected from January 2014 to December 2020 at Union Hospital, Tongji Medical College, Huazhong University of Science and Technology. This hospital is a large tertiary academic center performing over 5000 cardiac procedures annually, 98% of which are non-emergent [13].

Eligible patients in the Union cohort were stratified into the TCI group and the MCI group based on the anesthesia delivery method. In the MCI group, the induction regimen was performed at the attending

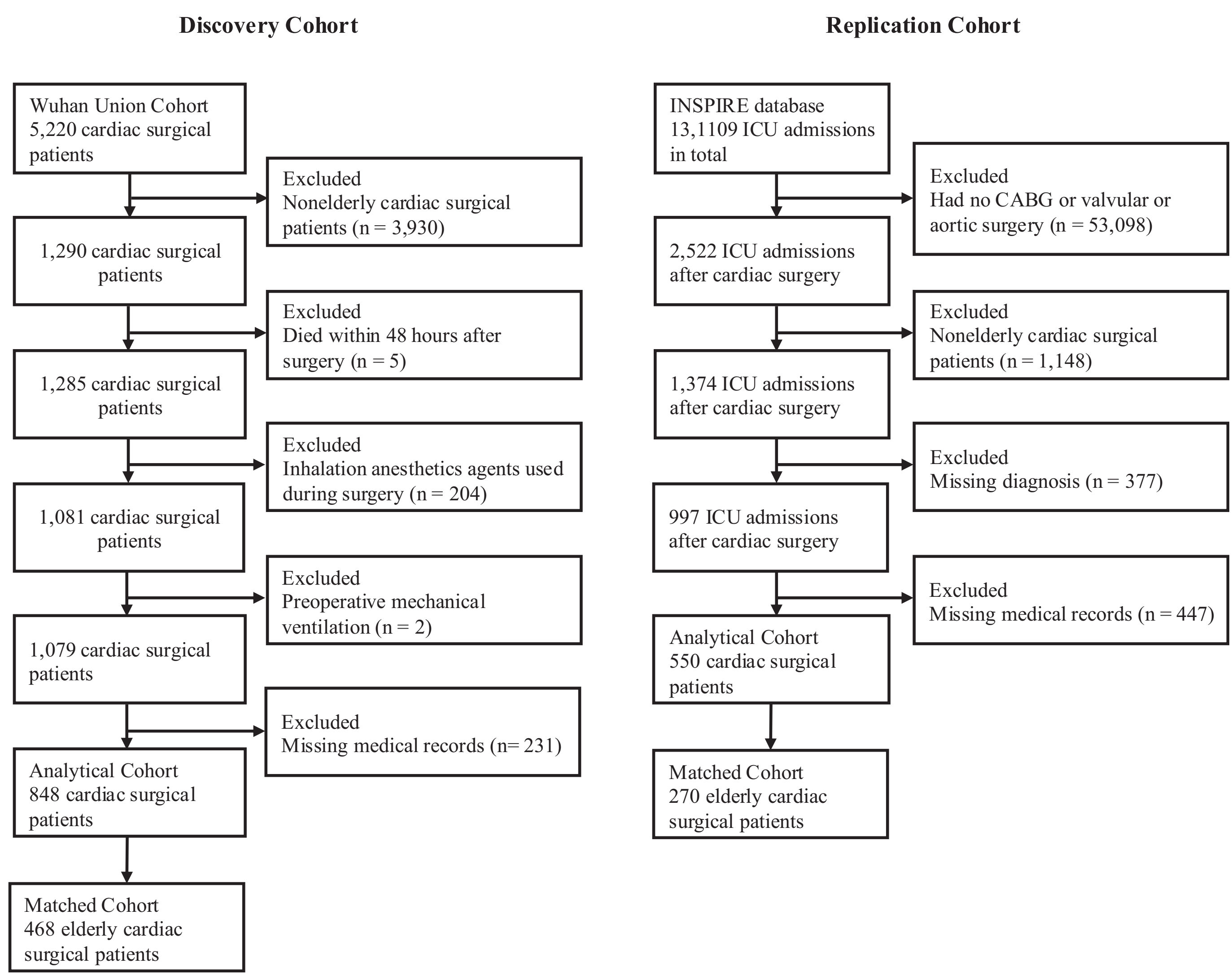

Fig. 1. Flow diagram of the selection process of the eligible population.
Abbreviations: CABG, coronary artery bypass grafting; INSPIRE, informative surgical patient dataset for innovative research environment

anesthesiologist's discretion. Following induction, anesthesia was maintained with manually controlled infusions of propofol and remifentanil. In the TCI group, anesthesia was induced and maintained using target-controlled infusion (TCI) systems. The pharmacokinetic models applied were the Marsh model for propofol and the Minto model for remifentanil [14,15]. Induction was achieved by gradually increasing the target plasma concentration of propofol until loss of consciousness, defined as the absence of response to tactile stimulation and corresponding to a score of 1 on the Observer's Assessment of Alertness/Sedation (OAA/S) Scale [16]. To ensure adequate anesthesia depth and prevent intraoperative awareness or excessive sedation, all patients were monitored using a Narcotrend Monitor (Monitor Technik, Bad Bramstedt, Germany), with the index maintained at the appropriate E2-D1 level. Vasoactive drugs were administered as per anesthetist's discretion. After surgery, all patients were transferred to the cardiac surgery ICU with mechanical ventilation and received standard post-cardiac surgery monitoring and treatment.

We further extended our analysis to enhance the generalizability of our findings by enrolling patients from the INSPIRE dataset, an international research dataset in perioperative medicine, which includes approximately 130,000 cases who underwent anesthesia for surgery at an academic institution in South Korea between 2011 and 2020 [17]. The same inclusion and exclusion criteria were applied to the INSPIRE dataset.

This study was approved by the Ethics Committee of Union Hospital (No. 2021–1070) and by the Institutional Review Board of Seoul National University Hospital (H-2210-078-1368). Considering the observational nature of this cohort study, the Ethics Committee of both institutions agreed to waive written informed consent. This study was reported according to STROBE guidelines [18]. A completed STROBE checklist has been provided as a separate Supplementary file to document compliance.

2.2. Study outcomes

The primary outcome is the length of ICU stays post-surgery. For patients with multiple ICU admissions after the same surgery, the length of ICU stays is calculated as the total days spent in the ICU.

Secondary outcomes include duration of mechanical ventilation, prolonged mechanical ventilation (PMV), reintubation, acute kidney injury (AKI), use of continuous renal replacement therapy (CRRT), and in-hospital mortality. PMV was defined as postoperative mechanical ventilation lasting more than 48 h or re-intubation within the first 2 days after surgery [19]. AKI was defined according to the serum creatinine criteria of the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines [20].

Perioperative clinical information for the enrolled patients was collected. Baseline characteristics included age, gender, body mass index (BMI), history of smoking and alcohol use, ejection fraction (EF), hypertension, diabetes, coronary artery disease (CAD), myocardial infarction (MI), previous percutaneous coronary intervention (PCI), stroke, renal insufficiency, recent pneumonia, and history of cardiac surgery. Intraoperative data included the anesthetic method, surgery type, surgery duration, CPB time, aortic cross-clamp (ACC) time, heart rate, and mean arterial pressure (MAP; recorded at 5-min intervals).

2.3. Intraoperative blood pressure monitoring

To further investigate the impact of the TCI versus MCI on hemodynamic stability, we quantified intraoperative blood pressure using “blood pressure variability” and “area under a MAP threshold”. All patients in both cohorts underwent intra-arterial blood pressure monitoring during the intraoperative period; data were documented at 5-min intervals and stored in an electronic database.

Blood pressure variability was assessed using two complementary indices: the variability independent of the mean (VIM) and the average real variability (ARV), calculated as follows:

$$ARV = \frac{1}{N-1}\sum_{i=1}^{N-1}|BP_{i+1} - BP_i|$$

where N denotes the number of valid BP measurements of a given subject, and BP_i represents the i^{th} BP reading.

$$VIM = SD \times \left(\frac{\overline{x}_{pop}}{\overline{x_i}}\right)^{\beta}$$

where β derived from the regression model $\ln SD = \alpha + \beta \times \ln(\overline{x_i})$. Here, $\overline{x_i}$ denotes the mean BP of a given subject, and, and $\overline{x}_{pop}$ denotes the mean BP of the study population.

VIM was calculated to quantify the dispersion of MAP values while minimizing dependence on the mean level, whereas the ARV reflected short-term fluctuations in MAP by averaging the absolute differences between consecutive measurements.

To further evaluate hypotension burden, the area under a MAP threshold was computed. The area was defined as the integral between the recorded MAP curve and a predefined MAP threshold over time. Specifically, the threshold was set at 65 mmHg during the non-CPB period and 50 mmHg during CPB, in accordance with guideline statements. [21,22]

2.4. Statistical analysis

Given the evident heterogeneity in participant characteristics between the Union and INSPIRE cohorts, a cohort-specific analysis was performed. All eligible patients were included in the analyses, and no a priori sample size calculation was conducted. To address concerns regarding statistical power, a post hoc power analysis was conducted using G*Power software based on the primary outcome of ICU length of stay, indicating an achieved statistical power of 0.84 (Supplementary Methods) [23,24].

For baseline characteristics, continuous variables were reported as median (IQR) and compared with the Mann–Whitney *U* test. Categorical variables were reported as counts (%) and assessed using the chi-square or Fisher's exact test.

To address potential confounding, propensity score matching was conducted. A logistic regression model was utilized to estimate the propensity score using the following clinically relevant variables: age, gender, BMI, hypertension, diabetes, CAD, surgery duration, and CPB time. The nearest-neighbor matching method was applied to achieve a 1:1 match with a caliper width of 0.02, as previously described [25]. Covariate balance after matching was evaluated using the standardized mean difference (SMD); an SMD value below 0.1 was considered to indicate a well-balanced distribution [26].

In the matched cohort, continuous outcomes were compared using linear regression models and reported as mean differences (MD) with 95% confidence intervals (CIs). Binary outcomes were analyzed using logistic regression models to estimate odds ratios (ORs) with 95% CIs. We further stratified the patients into four groups based on their ICU length of stay and calculated the proportion of patients in the TCI and MCI groups within each category. Forest plots were further generated to visualize the comparative effects of TCI and MCI across all outcomes.

To explore potential variations in the primary outcome, subgroup analyses were conducted across the following categories: gender (male or female), BMI ($<$28.0, or $\geq$ 28.0) and ASA Class. The *P*-value for interaction was calculated using a multivariable linear regression model. To explore potential mechanisms, ARV, VIM, and the area under the MAP threshold were compared between TCI and MCI in the propensity-matched cohorts using the Mann-Whitney *U* test.

Three sensitivity analyses were performed to assess the robustness of our findings to potential unmeasured confounding. First, to evaluate the potential impact of postoperative hemodynamic management on ICU length of stay and outcomes, we conducted an additional analysis in the INSPIRE cohort, which contains high-resolution postoperative monitoring data. Within the matched cohort, ARV, VIM and the area under the curve for MAP <65 mmHg during the first postoperative day in the ICU were compared between the TCI and MCI groups. Second, to minimize potential confounding related to the COVID-19 pandemic, we restricted the analysis to patients who underwent surgery between 2014 and 2019, excluding all cases from 2020. Propensity score matching was re-performed in this pre-pandemic sub-cohort, and primary and secondary outcomes were re-evaluated. Finally, to account for potential temporal variations in surgical techniques, anesthetic management, and medication use over the 6-year study period, the surgical year was incorporated as a covariate in the propensity score model. The study period was stratified into three intervals (2014–2015, 2016–2018, and 2019–2020), and comparative effects of TCI and MCI across all outcomes were re-assessed in this year-adjusted matched cohort.

Statistical analyses were performed using R version 4.5 (*R* Foundation, Vienna, Austria). Statistical significance was set at a two-tailed *p*-value of <0.05.

3. Results

3.1. Primary analyses in the union cohort

In the Union cohort, 848 elderly patients undergoing cardiac surgery were included (Fig. 1). Baseline and intraoperative characteristics are summarized in Table 1. Patients in the MCI group had a higher prevalence of hypertension (47.3% vs 33.8%; $P < 0.001$) and coronary artery disease (50.4% vs 37.8%; $P = 0.001$), underwent more complex surgery ($P < 0.001$), and had slightly shorter cardiopulmonary bypass times (111 vs 117 min, $P = 0.015$) than those in the TCI group.

In the unmatched cohort, the TCI group had a shorter ICU stay than the MCI group (mean 4.67 [SD 5.53] days vs 5.5 [SD 5.14] days; $P < 0.001$). The TCI group also had a lower incidence of prolonged mechanical ventilation (12.7% vs 27.6%; $P < 0.001$) and a shorter duration of mechanical ventilation (mean 22.8 [SD 26.8] h vs 30.3 [SD 56.7] h; $P < 0.001$) than the MCI group. Patients with an ICU stay of >2 days exhibited markedly higher rates of postoperative complications and mortality compared with those with shorter ICU stays (Supplementary Table S1).

Propensity score matching yielded 234 patients in each group with well-balanced demographic and clinical characteristics (Table 1). In the matched cohort, the TCI group had a significantly shorter mean length of ICU stay compared to the MCI group (Mean Difference [MD] -0.92 days, 95% CI: −1.68 to −0.17; $P = 0.017$). When patients were stratified by ICU stay duration, the proportion of MCI patients increased progressively across longer ICU-stay categories (Supplementary Fig. S1).

In the analysis of postoperative recovery metrics, TCI was associated with a shorter duration of postoperative mechanical ventilation (MD -11.6 h, 95% CI: −17.1 to −6.14; $P < 0.001$), and lower odds of reintubation (OR 0.94, 95% CI 0.90–0.99), acute kidney injury (OR 0.84, 95% CI 0.78–0.91), and in-hospital mortality (OR 0.95, 95% CI 0.92–0.99) (Fig. 2). No significant differences were observed in postoperative respiratory infections or the requirement for continuous renal replacement therapy (CRRT).

3.2. Extended analyses in the INSPIRE cohort

We replicated our analyses in the INSPIRE cohort, which included 550 eligible patients (53.5% male; mean age: 71.95 [SD 5.73] years). Baseline characteristics are summarized in Supplementary Table S2. After propensity score matching, 135 patients were retained in each study arm with adequately balanced covariates.

In the matched INSPIRE cohort, the TCI group had a significantly shorter mean length of ICU stay compared to the MCI group (MD -1.20 days, 95% CI: −2.32 to −0.80; $P = 0.036$). Stratification by ICU stay again demonstrated an increasing proportion of MCI patients at longer ICU durations, with only MCI patients represented among those with ICU stay ≥21 days (Supplementary Fig. S1).

Regarding secondary postoperative outcomes, TCI was associated with a lower rate of reintubation (OR 0.96, 95% CI 0.92–0.99), reduced requirement for CRRT (OR 0.90, 95% CI 0.85–0.94). Rates of respiratory infections, acute kidney injury, and in-hospital mortality were comparable between groups (Fig. 2).

Table 1
Baseline characteristics before and after PSM in the Union cohort.

	Full Cohort (n = 848)				Matched Cohort (n = 468)			
	TCI group (275)	MCI group(573)	*P* value	SMD	TCI group (234)	MCI group (234)	*P* value	SMD
Age (years)	68 (66,70)	68 (66,70)	0.622	−0.087	68 (66,70)	68 (66,70)	0.908	−0.037
Male	169(61.5)	322(56.2)	0.146	0.108	139 (57.7)	137 (56.8)	0.854	0.017
BMI (kg/m^2)	23.2 (21.5, 26.0)	23.4(21.3, 25.6)	0.844	−0.023	23.3 (21.3, 26.0)	23.4(21.4, 25.5)	0.956	−0.012
Smoking history	80 (29.1)	168 (29.3)	0.945	−0.005	72 (29.9)	62 (25.7)	0.309	0.091
Drinking history	61 (22.2)	136 (23.7)	0.616	−0.037	58 (24.1)	50 (20.7)	0.382	0.080
EF (%)	60 (57, 65)	61 (57, 65)	0.257	−0.073	60 (57, 65)	60 (56, 65)	0.975	−0.001
Hypertension	93 (33.8)	271(47.3)	<0.001	−0.285	87 (36.1)	88 (36.5)	0.925	−0.009
Diabetes	37 (13.5)	81(14.1)	0.788	−0.020	32 (13.3)	42 (17.4)	0.206	−0.122
Pneumonia	12 (4.4)	23(4.0)	0.811	0.017	9 (3.7)	10 (4.1)	0.815	−0.020
CAD	104 (37.8)	289(50.4)	0.001	−0.260	93 (38.6)	96 (39.8)	0.780	−0.014
History of MI	10 (3.6)	29 (5.1)	0.354	−0.076	9 (3.7)	12 (5.0)	0.503	−0.014
History of PCI	13 (4.7)	22 (3.8)	0.543	0.042	9 (3.7)	11 (4.6)	0.648	−0.039
Stroke	20 (7.3)	39 (6.8)	0.803	0.018	18 (7.5)	12 (5.0)	0.258	0.096
Renal insufficiency	41(14.9)	95 (16.6)	0.535	−0.047	35 (14.5)	38 (15.8)	0.703	−0.035
Prior cardiac surgery	7(2.5)	16 (2.8)	0.836	−0.016	6 (2.5)	7 (2.9)	0.779	−0.026
Type of surgery			0.027				0.962	
CABG only	50 (18.2)	85(14.8)		0.087	39 (16.2)	43 (17.8)		−0.043
Valve only	147 (53.5)	293(51.1)		0.047	136 (56.4)	129 (53.5)		0.058
Combined	78 (28.4)	195 (34.0)		0.136	11 (4.6)	10 (4.1)		0.016
Surgery time (h)	4.5 (3.6, 5.5)	4.2 (3.5, 5.3)	0.131	0.083	4.3 (3.5, 5.3)	4.3 (3.5, 5.3)	0.920	−0.032
CPB time (min)	117 (92,151)	111 (84,142)	0.015	0.114	116 (91, 148)	113 (88,143)	0.634	−0.029
ACC time (min)	78 (57, 100)	70 (50,93)	0.018	0.140	76 (55, 98)	73 (51,95)	0.490	−0.016

PSM = propensity score matching; MCI = manual controlled infusion; TCI = target-controlled infusion; SMD = standardized mean difference; BMI = body mass index; EF = ejection fraction; CAD = coronary artery disease; MI = myocardial infarction; PCI = percutaneous coronary intervention; CABG = coronary artery bypass grafting; CPB = cardiopulmonary bypass; ACC = aortic cross clamp.

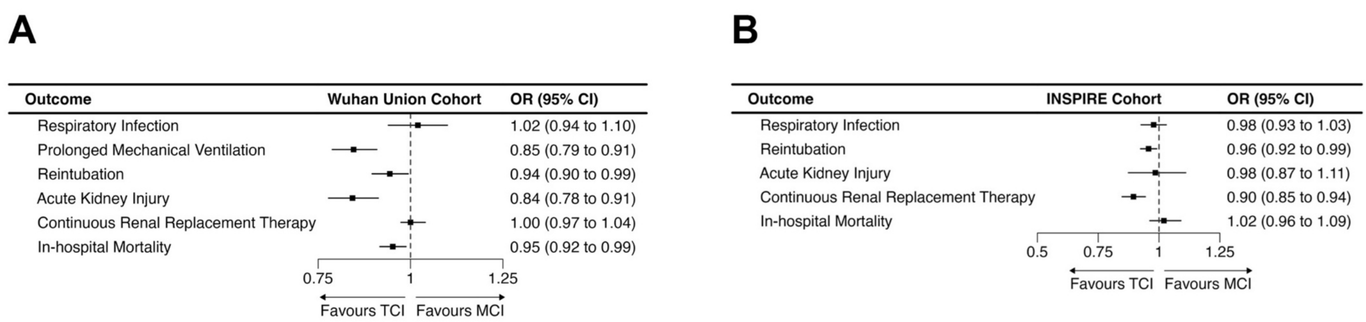

Fig. 2. (A) Forest plot of secondary outcomes in the Union Cohort. (B) Forest plot of secondary outcomes in the INSPIRE Cohort.
Abbreviations: ICU, intensive care unit; TCI, target-controlled infusion; MCI, manual-controlled infusion; OR, odds ratio

3.3. Intraoperative hemodynamic stability

As demonstrated in Table 2, the proportion of patients experiencing at least one intraoperative hypotensive episode did not differ between TCI group and MCI group in either cohort. However, both cohorts demonstrated consistently lower VIM and ARV values in the TCI group. Additionally, patients receiving TCI had significantly smaller areas under the MAP threshold of 65 mmHg compared with those receiving MCI (Fig. 3), indicating enhanced hemodynamic stability during surgery.

Supplementary Fig. S2 further characterizes the density distributions of MAP values for both cohorts. The INSPIRE cohort showed a left-shifted yet more concentrated MAP distribution, whereas the Union cohort exhibited a substantially wider distribution, reflecting greater dispersion of MAP values. This distributional pattern is consistent with the findings in Table 2, namely a higher incidence of hypotension but lower VIM and ARV indices in the INSPIRE cohort compared with the Union cohort.

3.4. Subgroup analyses

Subgroup analysis suggested the association between TCI and shorter ICU length of stay was stronger in patients with lower BMI in both cohorts (Union cohort: MD -1.22, 95% CI -2.34 to −0.1; INSPIRE cohort: MD -2.00, 95% CI -3.62 to −0.37). No other significant interaction effects were identified across the remaining prespecified subgroups (Fig. 4).

Table 2
Intraoperative hemodynamic parameters in the matched cohorts.

Variable	Union Cohort		P value	INSPIRE Cohort		P value
	TCI Group	MCI Group		TCI Group	MCI Group	
Hypotension incidence, n (%)	151 (64.5%)	138 (59.0%)	0.254	126 (93.3%)	128 (94.8%)	0.797
VIM of MAP, median [IQR]	0.304 [0.261, 0.355]	0.352 [0.281, 0.420]	<0.001	0.198 [0.182, 0.226]	0.205 [0.180, 0.237]	0.41
ARV of MAP, median [IQR]	15.7 [13.1, 18.2]	18.6 [15.0, 21.8]	<0.001	5.25 [4.20, 6.65]	5.94 [4.97, 7.51]	0.002
AUC < 65 mmHg, median [IQR]	270 [105, 612]	610 [381, 1158]	<0.001	370 [178, 840]	490 [215, 1025]	0.02

Data are presented as n (%) or median [IQR]. P values were calculated using the Chi-square test for hypotension incidence and the Mann-Whitney U test for continuous variables. Abbreviations: TCI, target-controlled infusion; MCI, manually controlled infusion; MAP, mean arterial pressure; VIM, variability independent of the mean; ARV, average real variability; AUC, area under the curve; IQR, interquartile range.

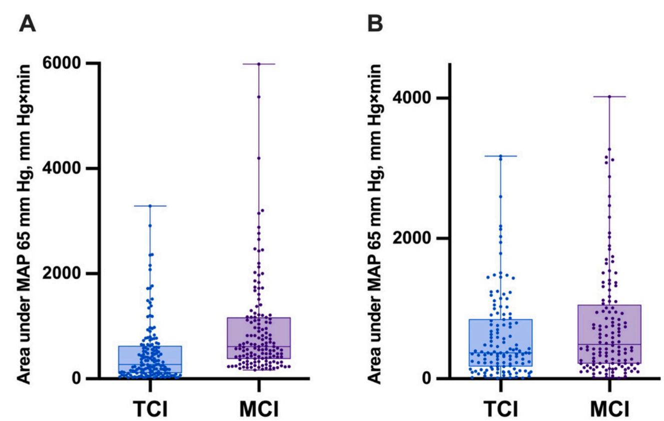

Fig. 3. (A) Areas under MAP values of 65 mmHg in the Union Cohort. (B) Areas under MAP values of 65 mmHg in the INSPIRE Cohort. Boxes represent the 25th and 75th quantiles; horizontal bold lines inside boxes, medians; Areas under MAP thresholds quantify both how long and to what extent MAP was below a respective threshold.
Abbreviations: MAP, mean arterial pressure

3.5. Sensitivity Analyses

To assess the potential confounding effect of postoperative hemodynamic management, we compared the hemodynamic PROFILES during the first 24 h after ICU admission within the matched INSPIRE cohort. No statistically significant differences were observed between the TCI and MCI groups in ARV, VIM, or the Area Under the Curve for MAP <65 mmHg (Supplementary Table S3). In a pre-pandemic sensitivity analysis restricted to patients treated between 2014 and 2019, TCI remained associated with a significantly shorter ICU length of stay after re-matching (Supplementary Table S4), while secondary outcomes showed consistent but non-significant trends (Supplementary Table S5). In the sensitivity analysis adjusting for surgical year, a clear temporal shift in anesthetic practice was observed, with TCI becoming predominant in later years (Supplementary Table S6). Accordingly, exact matching on surgical year resulted in a reduced sample size ($n = 73$ per group). Despite this reduction, TCI remained associated with a significantly shorter ICU length of stay (MD −1.67 days; 95% CI: −3.27 to −0.07; $P = 0.040$) (Supplementary Table S7), as well as lower rates of reintubation ($P = 0.020$) and in-hospital mortality ($P = 0.043$) (Supplementary Table S8).

4. Discussion

In this international multi-cohort retrospective study, we compared the effects of TCI versus MCI of propofol and remifentanil on postoperative recovery profiles in elderly patients undergoing cardiac

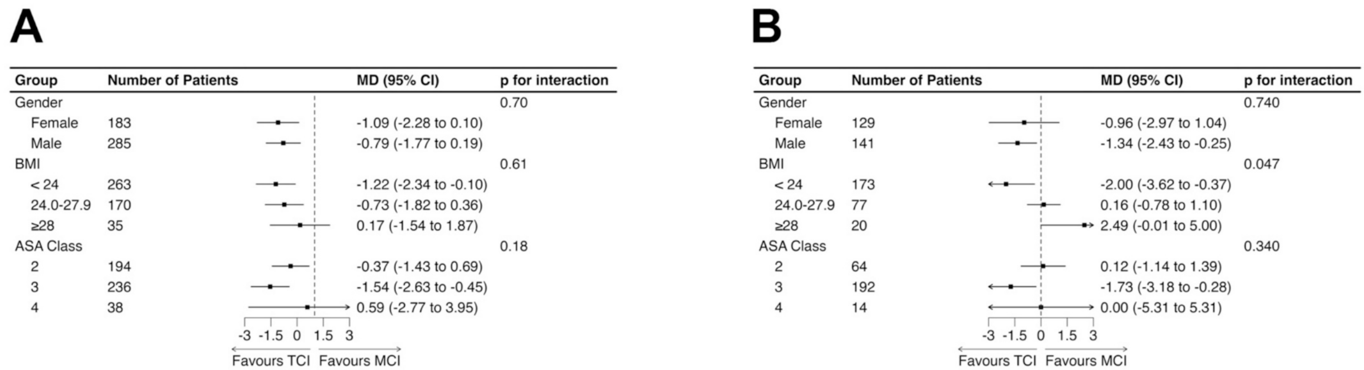

Fig. 4. Forest plot of odd ratio for subgroups of gender, BMI, and ASA class.
Abbreviations: OR, odds ratio; BMI, body mass index; ASA, American Society of Anesthesiologists

surgery. Our primary analysis revealed that TCI was associated with a significantly shorter length of ICU stay, a reduced incidence of prolonged mechanical ventilation, and lower reintubation rates compared with MCI. Additionally, TCI demonstrated superior intraoperative hemodynamic stability and potential kidney-protective effects. These findings were validated in an independent international cohort, underscoring the robustness and external validity of the results.

A key finding of our study is the improved intraoperative hemodynamic stability associated with TCI. In both cohorts, patients receiving TCI exhibited consistently lower variability in blood pressure and a reduced burden of hypotension, as indicated by lower VIM and ARV values and a smaller area under the curve for MAP <65 mmHg. Intraoperative hypotension and high blood pressure variability are well-established risk factors for postoperative organ dysfunction, including AKI and myocardial injury, particularly in elderly patients with reduced physiological reserve [11,12]. By targeting specific plasma or effect-site concentrations through pharmacokinetic modelling, TCI enables rapid, precise titration in response to changing surgical stimuli [7,27,28]. This likely prevents abrupt shifts in anesthetic depth and mitigates the hemodynamic swings more commonly observed with manual boluses or stepwise adjustments. Such stability reduces the risk of end-organ hypoperfusion, providing a plausible physiological mechanism underlying the lower AKI incidence and shorter ICU stay observed in the TCI group. The two cohorts exhibited distinct hemodynamic phenotypes, with the Union cohort showing greater MAP lability and the INSPIRE cohort demonstrating a more stable profile marked by higher hypotension incidence but lower variability indices. This likely reflects institutional differences in post-CPB management, where lower MAP targets are permissively maintained to reduce afterload, resulting in more frequent MAP values below 65 mmHg but with limited fluctuation.

Furthermore, the precise control offered by TCI also appears to contribute to improved respiratory outcomes. We observed a lower incidence of reintubation and prolonged mechanical ventilation in the TCI group. Postoperative respiratory failure and the need for reintubation are well-recognized determinants of prolonged ICU stay and increased mortality [29–31]. TCI systems mitigate the risk of drug accumulation and inadequate anesthesia depth by adjusting infusion rates based on patient-specific covariates such as age, gender, and weight [32]. In our primary cohort, depth-of-anesthesia monitoring (Narcotrend and OAA/S) provided additional control over inter-individual pharmacokinetic variability [33,34]. By avoiding relative overdose, TCI may facilitate emergence, quicker return of spontaneous ventilation, and better pulmonary recovery. Although we lacked data on total drug dosage, previous studies suggest TCI optimizes drug delivery efficiency and is associated with faster recovery time [6,35,36], which aligns with our findings. In our study, the reduction in ICU length of stay associated with TCI appears to be multifactorial. Despite similar rates of hypotension, TCI reduced both the cumulative burden and variability of hypotension, suggesting mitigation of hemodynamic instability rather than elimination of hypotensive events. Non-hemodynamic mechanisms, such as improved pharmacokinetic control and lower reintubation rates, may further contribute.

To our knowledge, this is the first study to specifically investigate the impact of TCI on comprehensive postoperative recovery profiles in cardiac surgery, extending beyond the scope of previous research which focused primarily on intraoperative hemodynamics or pharmacokinetic models [8–10]. The consistency of our findings across two diverse cohorts from different healthcare systems strengthens the generalizability of the results. While the reduction in ICU length of stay—approximately one day in the matched cohorts—is a statistically significant finding, it also carries clinical and economic relevance, potentially optimizing resource utilization in cardiac intensive care units.

Our study has several limitations. First, although propensity score matching reduces confounding, residual bias from unmeasured variables cannot be excluded. Second, due to the retrospective design, detailed data on intraoperative total drug dosage and precise timing of infusion adjustments were unavailable, preventing a direct comparison of drug consumption. Third, while both AKI and CRRT outcomes suggested a renal benefit of TCI, statistical significance differed between cohorts, likely reflecting differences in baseline characteristics, local CRRT initiation thresholds, and sample size. However, the hemodynamic data supporting greater stability in the TCI group were consistent across both cohorts, providing a mechanistic basis for the observed renal benefits. Fourth, although propensity score matching was used to balance measured baseline covariates and reduce selection bias, residual confounding from unmeasured or unknown factors cannot be excluded. Therefore, causal inferences should be interpreted with caution and cannot fully substitute for evidence from randomized controlled trials. Finally, incomplete ventilation data in the INSPIRE dataset limited our ability to fully validate the findings regarding prolonged mechanical ventilation, although the reduction in reintubation rates remained a robust and consistent signal.

5. Conclusion

This international multi-cohort study demonstrates that the use of TCI for total intravenous anesthesia is associated with superior intraoperative hemodynamic stability, shorter ICU stays, and improved postoperative recovery profiles in elderly patients undergoing cardiac surgery compared with MCI. These benefits were consistent across two independent cohorts. Implementing TCI in this high-risk population represents a simple, scalable process-of-care modification that may translate into meaningful improvements in clinical outcomes.

CRediT authorship contribution statement

Yun-Xiao Bai: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xia Li:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Shi-Qiang Chen:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Zhen-Zhen Xu:** Writing – original draft, Investigation, Formal analysis. **Yan-Ting Wang:** Writing – original draft, Formal analysis. **Fang-Ling Zhou:** Writing – original draft, Formal analysis. **Qing-Ping Wu:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Funding

This study was supported by the National Key Research and Development Program of China (Grant No. 2023YFC2506901).

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclinane.2026.112161.

References

[1] Sun LY, Jones PM, Wijeysundera DN, Mamas MA, Bader Eddeen A, O'Connor J. Association between handover of anesthesiology care and 1-year mortality among adults undergoing cardiac surgery. JAMA Netw Open 2022;5:e2148161.
[2] Lettino M, Mascherbauer J, Nordaby M, Ziegler A, Collet JP, Derumeaux G, et al. Cardiovascular disease in the elderly: proceedings of the European Society of Cardiology-Cardiovascular Round Table. Eur J Prev Cardiol 2022;29:1412–24.
[3] He D, Wang Z, Li J, Yu K, He Y, He X, et al. Changes in frailty and incident cardiovascular disease in three prospective cohorts. Eur Heart J 2024;45:1058–68.
[4] Engelman DT, Ben Ali W, Williams JB, Perrault LP, Reddy VS, Arora RC, et al. Guidelines for perioperative care in cardiac surgery: enhanced recovery after surgery society recommendations. JAMA Surg 2019;154:755–66.
[5] Alvarez-Nebreda ML, Bentov N, Urman RD, Setia S, Huang JC, Pfeifer K, et al. Recommendations for preoperative management of frailty from the society for perioperative assessment and quality improvement (SPAQI). J Clin Anesth 2018;47:33–42.
[6] Egan TD, Minto CF, Schnider TW. Steady-state trumps accuracy: target-controlled infusion as a gain switch. Br J Anaesth 2024;133:726–9.
[7] Absalom AR, Glen JI, Zwart GJ, Schnider TW, Struys MM. Target-controlled infusion: a mature technology. Anesth Analg 2016;122:70–8.
[8] Coetzee JF, Glen JB, Wium CA, Boshoff L. Pharmacokinetic model selection for target controlled infusions of propofol. Assessment of three parameter sets. Anesthesiology 1995;82:1328–45.
[9] Hüppe T, Maurer F, Sessler DI, Volk T, Kreuer S. Retrospective comparison of Eleveld, Marsh, and Schnider propofol pharmacokinetic models in 50 patients. Br J Anaesth 2020;124. e22-e4.
[10] Short TG, Hannam JA, Laurent S, Campbell D, Misur M, Merry AF, et al. Refining target-controlled infusion: an assessment of pharmacodynamic target-controlled infusion of propofol and remifentanil using a response surface model of their combined effects on bispectral index. Anesth Analg 2016;122:90–7.
[11] D'Amico F, Fominskiy EV, Turi S, Pruna A, Fresilli S, Triulzi M, et al. Intraoperative hypotension and postoperative outcomes: a meta-analysis of randomised trials. Br J Anaesth 2023;131:823–31.
[12] Sessler DI, Bloomstone JA, Aronson S, Berry C, Gan TJ, Kellum JA, et al. Perioperative quality initiative consensus statement on intraoperative blood pressure, risk and outcomes for elective surgery. Br J Anaesth 2019;122:563–74.
[13] Bai YX, Wang ZH, Lv Y, Liu J, Xu ZZ, Feng YQ, et al. Association between frailty and acute kidney injury after cardiac surgery: unraveling the moderation effect of body fat through an international, retrospective, multicohort study. Int J Surg 2025;111:761–70.
[14] Marsh B, White M, Morton N, Kenny GN. Pharmacokinetic model driven infusion of propofol in children. Br J Anaesth 1991;67:41–8.
[15] Minto CF, Schnider TW, Egan TD, Youngs E, Lemmens HJ, Gambus PL, et al. Influence of age and gender on the pharmacokinetics and pharmacodynamics of remifentanil. I model development. Anesthesiology 1997;86:10–23.
[16] Chernik DA, Gillings D, Laine H, Hendler J, Silver JM, Davidson AB, et al. Validity and reliability of the observer's assessment of alertness/sedation scale: study with intravenous midazolam. J Clin Psychopharmacol 1990;10:244–51.
[17] Lim L, Lee H, Jung CW, Sim D, Borrat X, Pollard TJ, et al. INSPIRE, a publicly available research dataset for perioperative medicine. Sci Data 2024;11:655.
[18] von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The strengthening the reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet 2007;370:1453–7.
[19] Siao SF, Ku SC, Tseng WH, Wei YC, Chang YC, Hsiao TY, et al. Effects of a swallowing and oral-care program on resuming oral feeding and reducing pneumonia in patients following endotracheal extubation: a randomized, open-label, controlled trial. Crit Care 2023;27:283.
[20] Kidney Disease: Improving gGlobal oOutcomes (KDIGO) aAcute kKidney iInjury wWork gGroup. KDIGO clinical practice guideline for acute kidney injury. Kidney Int Suppl 2012;2:1–138.
[21] Saugel B, Fletcher N, Gan TJ, Grocott MPW, Myles PS, Sessler DI. PeriOperative quality initiative (POQI) international consensus statement on perioperative arterial pressure management. Br J Anaesth 2024;133:264–76.
[22] Kunst G, Milojevic M, Boer C, De Somer F, Gudbjartsson T, van den Goor J, et al. 2019 EACTS/EACTA/EBCP guidelines on cardiopulmonary bypass in adult cardiac surgery. Br J Anaesth 2019;123:713–57.
[23] Faul F, Erdfelder E, Lang AG, Buchner A. G*power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. Behav Res Methods 2007;39:175–91.
[24] Faul F, Erdfelder E, Buchner A, Lang AG. Statistical power analyses using G*power 3.1: tests for correlation and regression analyses. Behav Res Methods 2009;41:1149–60.
[25] Kim HK, Jeon YJ, Um SW, Shin SH, Jeong BH, Lee K, et al. Role of invasive mediastinal nodal staging in survival outcomes of patients with non-small cell lung cancer and without radiologic lymph node metastasis: a retrospective cohort study. EClinicalMedicine 2024;69:102478.
[26] Austin PC. Balance diagnostics for comparing the distribution of baseline covariates between treatment groups in propensity-score matched samples. Stat Med 2009;28:3083–107.
[27] Passot S, Servin F, Pascal J, Charret F, Auboyer C, Molliex S. A comparison of target- and manually controlled infusion propofol and etomidate/desflurane anesthesia in elderly patients undergoing hip fracture surgery. Anesth Analg 2005;100:1338–42.
[28] Russell D. Intravenous anaesthesia: manual infusion schemes versus TCI systems. Anaesthesia 1998;53(Suppl. 1):42–5.
[29] Esteban A, Alía I, Gordo F, Fernández R, Solsona JF, Vallverdú I, et al. Extubation outcome after spontaneous breathing trials with T-tube or pressure support ventilation. The Spanish lung failure collaborative group. Am J Respir Crit Care Med 1997;156:459–65.
[30] Epstein SK, Ciubotaru RL, Wong JB. Effect of failed extubation on the outcome of mechanical ventilation. Chest 1997;112:186–92.
[31] Thille AW, Harrois A, Schortgen F, Brun-Buisson C, Brochard L. Outcomes of extubation failure in medical intensive care unit patients. Crit Care Med 2011;39:2612–8.
[32] Guarracino F, Lapolla F, Cariello C, Danella A, Doroni L, Baldassarri R, et al. Target controlled infusion: TCI. Minerva Anestesiol 2005;71:335–7.
[33] Schnider TW, Minto CF, Struys MM, Absalom AR. The safety of target-controlled infusions. Anesth Analg 2016;122:79–85.
[34] Nimmo AF, Absalom AR, Bagshaw O, Biswas A, Cook TM, Costello A, et al. Guidelines for the safe practice of total intravenous anaesthesia (TIVA): joint guidelines from the Association of Anaesthetists and the Society for Intravenous Anaesthesia. Anaesthesia 2019;74:211–24.
[35] Chiang MH, Wu SC, You CH, Wu KL, Chiu YC, Ma CW, et al. Target-controlled infusion vs. manually controlled infusion of propofol with alfentanil for bidirectional endoscopy: a randomized controlled trial. Endoscopy 2013;45:907–14.
[36] Wang X, Wang T, Tian Z, Brogan D, Li J, Ma Y. Asleep-awake-asleep regimen for epilepsy surgery: a prospective study of target-controlled infusion versus manually controlled infusion technique. J Clin Anesth 2016;32:92–100.